Kavach – Product Requirements Document

Hackathon: PS2 – Women's Safety | **Platform:** Flutter (Android-first demo) | **Deadline:** 25 Oct 2026, 9:00 PM IST **Deliverables:** GitHub repository link + demo video **Status:** v1.0 – ready for build

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1. One-line pitch

Kavach is a single safety app that protects a woman **before, during and after** an incident, across both the **physical world** (SOS, safe walks) and the **digital world** (harassment, blackmail, deepfakes), and turns every incident into **verifiable evidence** she can act on.

Tagline: *Prevent. Respond. Prove. Recover.*

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2. Problem statement (from the hackathon)

Women face safety challenges in physical and digital environments, from harassment and unsafe situations to privacy breaches, online threats and identity misuse. The solution must be effective, practical, and sensitive to safety and privacy.

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3. Research summary: what exists and what is missing

3.1 What existing apps already do (table stakes, not our differentiator)

SOS button, live location sharing to contacts, fake call, calling police (112), area safety ratings (Safetipin), nearby safe places.

3.2 Research gaps we address

#	Gap found in research	Evidence	How Kavach addresses it
1	**Physical and digital safety are fragmented.** Apps either send alerts or detect online abuse, rarely both.	SafeSpace paper (arXiv 2508.16488); our review of Indian apps	One app, one Evidence Vault, one complaint generator for both

#	Gap found in research	Evidence	How Kavach addresses it
2	**Deepfakes and image abuse are ignored by safety apps.** About 10% of cases on an Indian abuse helpline involve AI-manipulated images; platforms and law respond slowly.	Rati Foundation / Tattle report via The Guardian (Nov 2025)	Deepfake Shield + takedown/complaint workflow
3	**Almost no offline support.** Only 15% of 178 VAWG apps reviewed offered offline functionality.	JMIR Formative Research (Rahman et al.)	SMS fallback, offline-first local storage, queue-and-sync
4	**Reliability flaws.** Late messages, battery drain, background failures, recording malfunctions.	Masud et al. 2022, cited by Transport Scotland review	Foreground service, delivery confirmation, SOS Test Mode
5	**Single-trigger and false alarms.** Shake/press-only triggers fail in real situations and cause false alarms.	Atlantis Press review of women's safety systems; The News Minute	Multi-trigger (shake, hold, voice) with cancel window and PIN
6	**Surveillance and the digital leash.** Apps can become tools that control women's movement.	The News Minute	Consent-first, time-boxed, user-controlled sharing
7	**Low real-world use and no effectiveness evaluation.** 72% of surveyed women with safety apps had never used them.	Red Elephant 2016 (via The News Minute); Ford et al. 2022	Habit features (Safe Walk, check-ins), SOS Test Mode, delivery receipts
8	**Abuser is often known to the victim** (family, partner), so contacting family or police is not always safe.	Nayantara's research via The News Minute	Decoy mode, discreet UX, NGO and counselor routing, private evidence

#	Gap found in research	Evidence	How Kavach addresses it
9	**Language barrier.** Voice triggers and content mostly in English or Hindi.	EveSafe project; JMIR review	Kannada, Hindi, English UI and voice keywords
10	**Accessibility.** Many apps fail for visually impaired users.	Ranganathan 2017 (India study, 50 apps)	Large targets, screen-reader labels, haptic and voice feedback
11	**Weak public-sector link.** Few apps connect meaningfully to emergency services.	JMIR review; Maxwell et al. 2020	112 and 181 integration (call + structured alert), complaint PDFs for police and cyber cell

3.3 Novelty claim (state this carefully in the pitch)

We did not find an app that combines **(a)** multi-trigger SOS, **(b)** a hashed evidence vault covering both physical and digital incidents, **(c)** deepfake and abusive-message analysis, and **(d)** auto-generated complaint packets, with **(e)** a consent-first privacy model. Pitch it as *unified and evidence-first*, not as the world's first.

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4. Target users and personas

- **Ananya, 21, college student, Mangaluru.** Travels by bus, faces street harassment, receives abusive DMs. Prefers Kannada/English.
- **Priya, 28, working professional.** Late commutes, wants discreet safety. Worried about leaked or morphed photos.
- **Savitri, 40, lives with an unsafe family member.** Cannot openly use a safety app. Needs decoy mode and private evidence.
- **Trusted contact (Meera).** Receives alerts, needs a simple, reliable view. Does not need to install the app.

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5. Goals and non-goals

Goals

1. Send a reliable alert to trusted contacts within seconds, even offline (via SMS).
2. Capture tamper-evident evidence for physical and digital incidents.
3. Help victims of online abuse and deepfakes act (analyze, document, report).

4. Respect privacy: no permanent tracking, user controls everything.

Non-goals (for the hackathon)

- Real integration with police dispatch systems or the official 112 backend.
- iOS parity (Android-first; iOS limits background and silent SMS).
- Legal advice. The app gives information and templates only.
- Wearable hardware.

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6. Feature scope and priority

P0 = must ship for demo. P1 = should ship. P2 = stretch.

6.1 Account and onboarding (P0)

- Splash, 3-screen onboarding (Prevent, Respond, Prove), permissions explainer.
- Phone OTP login (Firebase Auth). Optional name and email.
- Profile: name, language, blood group (optional), emergency note.
- App lock: PIN or biometric.

6.2 Trusted Circle (P0)

- Add 1 to 5 contacts (pick from phone contacts or manual).
- Per contact: name, phone, relationship, notify-by (SMS / call).
- Contacts do not need the app. They get an SMS with a map link.

6.3 SOS engine (P0)

- **Triggers:** big in-app SOS button (press and hold 2s), shake gesture, voice keyword ("help" / "bachao" / "kaapadi"), lock-screen persistent notification button.
- **Countdown:** 5-second cancel window with PIN to prevent false alarms.
- **Actions on trigger:** SMS to all contacts with live Google Maps link; auto-dial first contact or 112 (user setting); start evidence capture; show SOS-active screen.
- **Offline fallback:** SMS works without internet; data queued and synced later.
- **SOS Test Mode:** sends a clearly marked test message so users can verify the chain works.
- **Delivery log:** shows which contacts were sent the alert and when.

6.4 Evidence Vault (P0, core differentiator)

- Auto-capture on SOS: audio (30 to 60 s), a photo, GPS point, device time.

- Manual add: screenshots, photos, notes (for online harassment).
- Every item gets a **SHA-256 hash and timestamp** stored in a case record; items are encrypted locally.
- **Cases:** group items into a case (e.g., "Messages from unknown number").
- **Export:** one-tap PDF "Incident Report" with timeline, hashes, location, and a plain-language summary.

6.5 Digital Safety Shield (P0 for text, P1 for image)

- **Message analyzer:** paste or share harassing text. Output: category (abuse, threat, sexual harassment, blackmail/extortion), severity, and recommended next step. Works for English, Hindi, Kannada (via multilingual model or LLM).
- **Deepfake check:** upload an image and receive a manipulation-likelihood score with a clear "this is an estimate" disclaimer. Backed by a FastAPI endpoint (team's own deepfake model).
- **Harassment Log:** timeline of incidents with repeat-offender pattern view (same sender or handle).
- **Action guide:** step-by-step: do not delete, save evidence, report on platform, file at cybercrime.gov.in, StopNCII.org for intimate-image hashing. Pre-filled complaint text.
- **Privacy promise:** images analyzed are not stored on the server.

6.6 Safe Walk (P1)

- User sets destination and expected duration. If she does not tap "I'm safe" before the timer ends, a check-in prompt appears; if unanswered, SOS fires automatically.
- Contacts get a start and arrival message.

6.7 Safety Map (P1)

- OpenStreetMap-based map (flutter_map, no API key needed).
- Crowd-sourced "unsafe spot" and "safe place" pins with category and time of day.
- Nearby police stations, hospitals (static seed data for demo).

6.8 Help and Recovery (P1)

- Helplines: 112 (emergency), 181 (women helpline), cybercrime portal link. One-tap call.
- Support directory: NGOs and counselors (seed data).
- Know-your-rights cards (short, plain language, English/Kannada/Hindi).

6.9 Privacy and Stealth (P1)

- **Decoy mode:** app opens as a calculator. Real PIN opens Kavach; **panic PIN** opens a fake empty vault and silently triggers SOS.
- **Consent-first sharing:** location sharing always time-boxed with auto-expiry; no background tracking outside SOS or Safe Walk.
- **Data controls:** export my data, delete everything, view what was shared.

6.10 Settings and accessibility (P1)

- Language switch (English, Kannada, Hindi).
- Trigger sensitivity, countdown length, auto-call preference.
- Large-touch mode, screen-reader labels, haptic and voice confirmation.

6.11 Stretch (P2)

- Live tracking web page for contacts (Firebase Hosting).
- Fake incoming call.
- Home-screen widget / quick tile for SOS.
- Pattern analytics for unsafe hotspots.

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7. Screen inventory and navigation

Bottom navigation (4 tabs): Home, Shield, Vault, Help. Settings via profile icon.

1. Splash
2. Onboarding (3 screens)
3. Login: phone number, OTP
4. Permissions explainer (location, mic, camera, SMS, notifications, contacts)
5. Profile setup
6. Trusted Circle: list, add, edit
7. **Home:** SOS hold button, status card ("Protection active"), quick actions (Safe Walk, Fake call, Test SOS)
8. SOS countdown and SOS active screen
9. Safe Walk: setup, active, check-in
10. **Shield:** tabs for Messages, Images, Harassment Log
11. Message analyzer: input, result
12. Deepfake check: upload, result
13. Action guide and complaint draft

14. **Vault:** cases list, case detail, add evidence, export PDF
15. Safety Map
16. **Help:** helplines, NGOs, rights cards
17. Settings: language, triggers, privacy, data controls, decoy and panic PIN
18. Decoy calculator

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8. Key user flows

Flow A – Emergency SOS Hold SOS (or shake) → 5 s countdown with cancel PIN → SMS and call fire → audio, photo, GPS captured and hashed into a new case → SOS-active screen shows delivery status → user can export report later.

Flow B – Harassing messages Shield → Messages → paste text → result (category, severity) → "Save to Vault" (hash and timestamp) → Action guide → export complaint draft for cybercrime.gov.in.

Flow C – Morphed image Shield → Images → upload → deepfake score → "Create case" → guided steps (platform report, StopNCII, cybercrime portal) → export packet.

Flow D – Unsafe household Open decoy calculator → enter real PIN for the app, or panic PIN to show an empty vault and alert contacts silently.

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9. Technical architecture

Client: Flutter 3.x, Dart, state management with Riverpod, navigation with go_router.

Backend: Firebase (Auth phone OTP, Firestore, Storage for optional encrypted backups, Cloud Messaging). A small **FastAPI** service for ML (deepfake and toxicity).

Key packages

- Auth and data: firebase_auth, cloud_firestore, firebase_core
- Location: geolocator
- Sensors: sensors_plus (shake)
- Background: flutter_foreground_task or flutter_background_service
- SMS: telephony-style plugin (Android silent SMS), url_launcher fallback
- Media: record, camera, image_picker
- Security: crypto (SHA-256), flutter_secure_storage, encrypt, local_auth
- Voice: speech_to_text
- Storage (local, offline): hive or isar
- PDF: pdf, printing

- Maps: flutter_map, latlong2
- Contacts: flutter_contacts
- Localization: flutter_localizations, intl

ML service (FastAPI)

- `POST /deepfake`: image in, `{score, label, note}` out. Uses the team's existing deepfake detection model.
- `POST /toxicity`: text in, `{category, severity, scores}` out. Use a multilingual toxicity model from Hugging Face or an LLM call with a strict JSON schema.
- Deploy on Render, Railway, or a Colab + tunnel for the demo. Keep a cached fallback so the demo never fails live.

Honest platform constraints (important for the build)

- Silent SMS and background services work reliably on **Android**. iOS blocks silent SMS; demo on Android.
- Flutter cannot detect power-button presses. Use shake, hold button, voice and notification actions instead.
- Shake and voice detection need a foreground service with a visible notification (Android requirement).
- ML scores are estimates. Show disclaimers; never present a score as proof.

Data model (Firestore + local)

- `users/{uid}`: name, phone, language, createdAt, settings
- `users/{uid}/contacts/{id}`: name, phone, relation, notifyBy
- `users/{uid}/cases/{caseId}`: title, type (physical, digital), createdAt, status
- `users/{uid}/cases/{caseId}/items/{itemId}`: type (audio, photo, screenshot, text, location), localPath, sha256, capturedAt, notes
- `users/{uid}/sos_events/{id}`: triggeredAt, trigger, location, deliveryLog[], isTest
- `reports/{id}`: lat, lng, category, timeOfDay, createdAt (anonymous safety pins)
- `ngos/{id}`: name, city, phone, focus (seed data)

Evidence files stay **on device (encrypted)** by default; only hashes and metadata sync.

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10. Non-functional requirements

- **Reliability:** SOS fires in under 3 seconds after countdown ends; works with no data connection (SMS).
- **Battery:** no continuous GPS; location only during SOS and Safe Walk.

- **Security:** local encryption, PIN/biometric lock, no ads, no third-party analytics, minimal permissions explained in plain language.
- **Privacy:** explicit consent screens, delete-all button, no selling or sharing of data.
- **Accessibility:** minimum 48 dp touch targets, semantic labels, high contrast, haptics.
- **Localization:** English, Kannada, Hindi for all P0 screens.

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11. Success metrics (for the pitch)

- Time from trigger to first alert sent (target: under 5 s including countdown cancel window)
- Alert delivery success in Test Mode (target: 100% on demo device)
- Time to produce an exportable evidence report (target: under 30 s)
- Message analyzer category accuracy on a small hand-labeled test set (report honestly)
- Percentage of onboarding users who add at least one trusted contact

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12. Risks and mitigations

Risk	Mitigation
Background SMS and shake detection fail on some phones	Test on one Android device early; show visible foreground notification; fall back to in-app button
ML endpoint down during demo	Cache sample results; record demo with a warm server
Scope too large for one day	Strict P0 first; cut decoy mode, then Safety Map, before cutting Vault or Shield
False alarms	Countdown plus PIN cancel; Test Mode
Overclaiming accuracy	Disclaimers; evaluate on a small labeled set and state results honestly
Privacy concerns about evidence	Local-first encryption; hashes only sync

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13. Build plan (until 25 Oct, 9 PM)

Tonight (24 Oct)

- Repo, Firebase project, theme, navigation, localization skeleton
- Login (OTP), onboarding, profile, Trusted Circle
- SOS: button, countdown, SMS with map link, call

25 Oct, morning

- Evidence Vault: capture, hash, case storage, PDF export
- Foreground service with shake trigger, SOS Test Mode, delivery log
- FastAPI: toxicity endpoint, deepfake endpoint deployed

25 Oct, afternoon

- Shield screens (message analyzer, deepfake check, action guide, complaint draft)
- Help screen with helplines and seed NGOs
- Safe Walk, then decoy mode if time remains

By 6 PM: feature freeze, bug fixes, README with screenshots **6 to 8 PM:** record demo video (2 to 3 min), upload **By 8:30 PM:** submit GitHub link and video. Keep a 30-minute buffer.

Suggested split for two people

- Person A (Flutter UI and flows): auth, onboarding, contacts, home, Vault UI, Help, localization
- Person B (engine and ML): SOS engine, background service, hashing and PDF export, FastAPI endpoints, Shield integration

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14. Demo video script (about 2.5 minutes)

1. **0:00 to 0:20** Problem: unsafe streets and online abuse, apps handle only one half.
2. **0:20 to 0:50** Onboarding, add trusted contact, show privacy-first permissions.
3. **0:50 to 1:25** SOS via shake: countdown, SMS received on second phone, evidence auto-captured, delivery log.
4. **1:25 to 1:55** Shield: paste a threatening message, see classification, save to Vault.
5. **1:55 to 2:15** Deepfake check on a sample image, guided action steps.
6. **2:15 to 2:35** Export the PDF incident report with hashes and timeline.
7. **2:35 to 2:50** Decoy mode (if built), closing line: *Prevent. Respond. Prove. Recover.*

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15. GitHub repository

Suggested repo name: `kavach-womens-safety-app` (Alternatives: `kavach-safety`, `kavach-flutter`)

Description: Unified physical and digital safety app for women: multi-trigger SOS, tamper-evident evidence vault, deepfake and harassment shield. Flutter + Firebase + FastAPI.

Topics: flutter, women-safety, hackathon, firebase, deepfake-detection, sos, fastapi, dart

Suggested structure

```
kavach-womens-safety-app/
  app/                 # Flutter project
    lib/
      core/            # theme, router, l10n, utils
      features/
        auth/
        contacts/
        sos/
        vault/
        shield/
        safe_walk/
        map/
        help/
        settings/
    assets/
    test/
  ml_service/          # FastAPI (deepfake, toxicity)
  docs/                # PRD, architecture, screenshots
  README.md
  LICENSE
  .gitignore
```

Git workflow (two people): `main` is always runnable. Work on `feat/<name>` branches, small PRs, merge often to avoid conflicts. Commit messages: `feat:`, `fix:`, `docs:`.

Setup commands

```
flutter create --org com.kavach --project-name kavach app
git init && git add . && git commit -m "chore: init project"
git branch -M main
git remote add origin <your-repo-url>
git push -u origin main
```

README must include: problem, solution, feature list, screenshots, architecture diagram, setup steps, tech stack, demo video link, team, limitations and disclaimers.

Do not commit: `google-services.json`, API keys, model weights over 100 MB (use a download link instead).

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16. Disclaimers to include in the app and README

- Kavach does not replace emergency services; call 112 in danger.
- Deepfake and toxicity results are probabilistic estimates, not legal proof.
- Complaint drafts are templates, not legal advice.

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17. References

- Rahman et al., Mobile Apps to Prevent Violence Against Women and Girls, JMIR Formative Research (178 apps reviewed; 15% offline)
- Transport Scotland, Personal safety apps: limitations and areas for further research
- The News Minute, Women's safety apps: without understanding gender violence, technology won't help
- SafeSpace: integrated digital safety platform, arXiv 2508.16488
- Rati Foundation / Tattle report on AI deepfakes in India, reported by The Guardian (Nov 2025)
- Atlantis Press, Offline-capable proactive emergency response in women's safety systems
- Indian app landscape: 112 India, My Safetipin, bSafe, Hawk Eye, I'M SAFE and others